ECLYPSE AI

RAG Evaluation Dataset

Engineering Assessment · Domain-specific Documents.

Overview

At Eclypse AI we process large, dense medical documents using LLM systems built on top of retrieval-augmented generation (RAG). For production use we expect to deploy on small local LLMs rather than frontier models.

To know whether a RAG system actually works on our material — and whether a new prompt, retrieval setting, or model swap helped or hurt performance — we need evaluation datasets that sit within the domain of the documents we process.

The assignment has two parts. The first part is building a system that constructs a meaningful evaluation dataset from a real document. The second part is demonstrating that the dataset is fit for purpose by using it to evaluate a small-LLM RAG.

The Corpus

You'll work over **NICE Technology Appraisal TA1063**. This is a NICE technology appraisal evidence collection. It is representative of the kind of dense, semi-structured material we work with on the proprietary side — without any of the data-handling constraints.

Source: https://www.nice.org.uk/guidance/ta1063/evidence/draft-guidance-consultation-committee-papers-pdf-15313389517

Part 1 — Build a Dataset-Construction System

The primary deliverable. Build a *system* (not just an output file) that produces an evaluation dataset for a RAG over TA1063.

Anything is fair game. You can call frontier LLMs (GPT-5-class, Claude-class, etc.) to propose candidate items, extract anchors, generate distractors, validate items, or suggest categories. You can use programmatic filters, embedding-similarity checks, retrieval-baseline sanity checks. The dataset construction should be a pipeline that an engineer at Eclypse could rerun on a different medical document next month and get a comparable result.

Deliverables for this part:

- **Dataset:** the evaluation dataset itself, in a structured machine-readable format (JSONL or similar).
- **Construction code:** the code that produced it — runnable end-to-end on the corpus.
- **Methodology writeup:** a short description of what your pipeline does and why.

Part 2 — Demonstrate the Dataset Works

Build a RAG over TA1063 using a **small local LLM** (Llama-3.2-3B, Phi-3.5-mini, Qwen2.5-3B, or similar) and run your dataset against it. This is the proof that your Part 1 work produced something useful — and a chance to find out where it didn't.

Tech Stack

Your choice. We have no preference on the rest of the stack: embedding model, vector store, chunking strategy, framework. Use what you'd use in production.

Submission

- **GitHub repo** (private link is fine) containing the construction code, the eval dataset, the RAG implementation, the eval harness, a README.md with run instructions, and your writeup.
- The **writeup** can live in the README or as a separate WRITEUP.md.